Question 1: What is hypothesis testing in statistics?
Answer Hypothesis Testing in statistics is a method used to make decisions or draw conclusions about a population based on sample data.

Hypothesis testing helps you check whether a claim about a population is likely to be true or not by using sample data.

Question 2: What is the null hypothesis, and how does it differ from the alternative hypothesis?
Answer The null hypothesis is the default assumption or claim that there is no effect, no difference, or no relationship in the population.

It represents the status quo.
We test data to see if we have enough evidence to **reject** it.

Difference between alternative hypothesis
1.There is an effect / difference
2.Claim to be tested

Question 3: Explain the significance level in hypothesis testing and its role in deciding the outcome of a test.
Answer The significance level, denoted by **α**, is the threshold used in hypothesis testing to decide whether to reject the null hypothesis.

It represents the maximum probability of making a Type I error (rejecting a true null hypothesis) that we are willing to accept.

Compare p-value with α
Controls the risk of false rejection

Question 4: What are Type I and Type II errors? Give examples of each.
Answer Type I Error : Rejecting the **null hypothesis (H_0)** when it is actually **true**.
Example A judge wrongly convicts an innocent person.

Type II Error Failing to reject the **null hypothesis (H_0)** when it is actually **false**.
Example A judge lets a guilty person go free.

Question 5: What is the difference between a Z-test and a T-test? Explain when to use each.
Answer: A **Z-test** and **T-test** are both hypothesis tests used to compare means, but they are used under different conditions.

1. Z-test

When to Use

- **Sample size is large** ($n \geq 30$)
- **Population standard deviation (σ) is known**
- Data is approximately normally distributed

2. T-test

When to Use

- **Sample size is small** ($n < 30$)
- **Population standard deviation (σ) is unknown**
- Data is approximately normally distributed

Question 6: Write a Python program to generate a binomial distribution with n=10 and p=0.5, then plot its histogram.

Answer

```
 import numpy as np
import matplotlib.pyplot as plt

# Parameters
n = 10      # number of trials
p = 0.5     # probability of success
```

```
size = 1000  # number of samples

# Generate binomial distribution
data = np.random.binomial(n, p, size)

# Plot histogram
plt.hist(data, bins=range(0, n+2), edgecolor='black')
plt.title("Binomial Distribution (n=10, p=0.5)")
plt.xlabel("Number of Successes")
plt.ylabel("Frequency")
plt.show()
```

Question 7: Implement hypothesis testing using Z-statistics for a sample dataset in Python. Show the Python code and interpret the results. sample_data = [49.1, 50.2, 51.0, 48.7, 50.5, 49.8, 50.3, 50.7, 50.2, 49.6, 50.1, 49.9, 50.8, 50.4, 48.9, 50.6, 50.0, 49.7, 50.2, 49.5, 50.1, 50.3, 50.4, 50.5, 50.0, 50.7, 49.3, 49.8, 50.2, 50.9, 50.3, 50.4, 50.0, 49.7, 50.5, 49.9]

Answer

```
import numpy as np
from scipy.stats import norm

sample_data = [49.1, 50.2, 51.0, 48.7, 50.5, 49.8, 50.3, 50.7, 50.2, 49.6,
               50.1, 49.9, 50.8, 50.4, 48.9, 50.6, 50.0, 49.7, 50.2, 49.5,
               50.1, 50.3, 50.4, 50.5, 50.0, 50.7, 49.3, 49.8, 50.2, 50.9,
               50.3, 50.4, 50.0, 49.7, 50.5, 49.9]

# Known population mean and std deviation
mu_0 = 50
sigma = 1   # assumed known

# Compute sample mean
x_bar = np.mean(sample_data)

# Sample size
n = len(sample_data)
```

```python
# Z-statistic
z = (x_bar - mu_0) / (sigma / np.sqrt(n))

# p-value (two-tailed)
p_value = 2 * (1 - norm.cdf(abs(z)))

print("Sample Mean:", x_bar)
print("Z-statistic:", z)
print("p-value:", p_value)
```

Question 8: Write a Python script to simulate data from a normal distribution and calculate the 95% confidence interval for its mean. Plot the data using Matplotlib.

Answer

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# 1. Simulate data from a normal distribution
np.random.seed(42)  # for reproducibility
mean = 50
std_dev = 10
n = 500

data = np.random.normal(mean, std_dev, n)

# 2. Calculate 95% Confidence Interval for the mean
sample_mean = np.mean(data)
sample_std = np.std(data, ddof=1)

# Standard Error
se = sample_std / np.sqrt(n)

# t-critical value for 95% CI
t_critical = stats.t.ppf(0.975, df=n-1)

# Confidence interval
```

```
lower_bound = sample_mean - t_critical * se
upper_bound = sample_mean + t_critical * se

print("Sample Mean:", sample_mean)
print("95% Confidence Interval: (", lower_bound, ",", upper_bound, ")")

# 3. Plot the data
plt.hist(data, bins=30, edgecolor='black')
plt.title("Simulated Normal Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```

Question 9: Write a Python function to calculate the Z-scores from a dataset and visualize the standardized data using a histogram. Explain what the Z-scores represent in terms of standard deviations from the mean.

Answer

```
import numpy as np
import matplotlib.pyplot as plt

# Function to calculate Z-scores
def calculate_z_scores(data):
    mean = np.mean(data)
    std = np.std(data, ddof=1)   # sample standard deviation
    z_scores = (data - mean) / std
    return z_scores

# Sample dataset
data = np.array([12, 15, 14, 10, 18, 20, 13, 16, 17, 19])

# Calculate Z-scores
z_scores = calculate_z_scores(data)

print("Z-scores:", z_scores)

# Plot histogram of standardized data
```

```
plt.hist(z_scores, bins=10, edgecolor='black')
plt.title("Histogram of Z-scores (Standardized Data)")
plt.xlabel("Z-score")
plt.ylabel("Frequency")
plt.show()
```